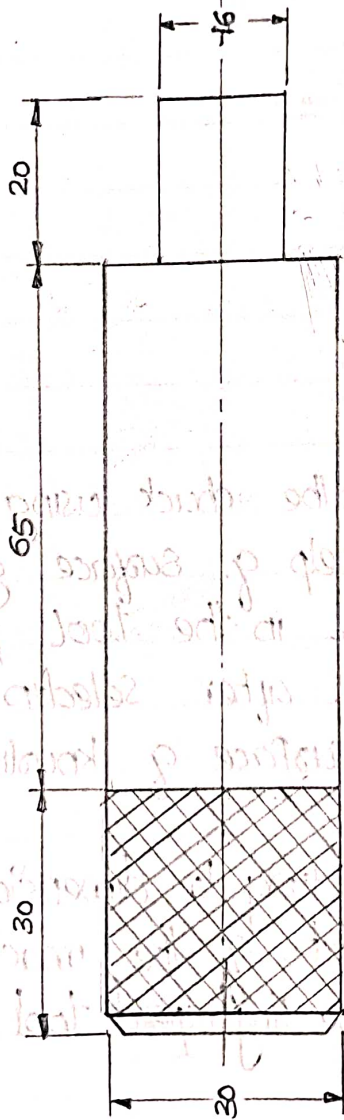


STEP TURNING



All dimensions are in "mm"

Scale: 1:1

STEP TURNING.AIM:-

To perform step turning operation on the given cylindrical workpiece.

MATERIAL REQUIRED:-

- (1) Lathe with standard accessories
- (2) Workpiece.

TOOLS REQUIRED

Chuck key, Cutting tool, (H.S.S) Vernier caliper, spanner etc.

PRINCIPLE

The cutting tool is adjusted for the desired depth of cut using the cross slide, thus on the workpiece rotates the ~~fast~~ tool is advanced slowly and relatively steady in a direction parallel to the rotational axis of the spindle. The motion is known as feed. The combined motion can ~~therefore~~ cause the workpiece by adjusted the feed so that the helical path of the tool overlaps and generates a cylindrical surface on the workpiece.

PROCEDURE

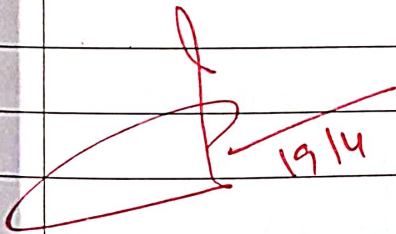
- (1) The given workpiece is fixed in chuck and centered.
- (2) The H.S tool bit positioned in the cutting tool is kept at

an angle to be the axis of the given workpiece. During this process positioned in the tool holder, the speed of the lathe is high.

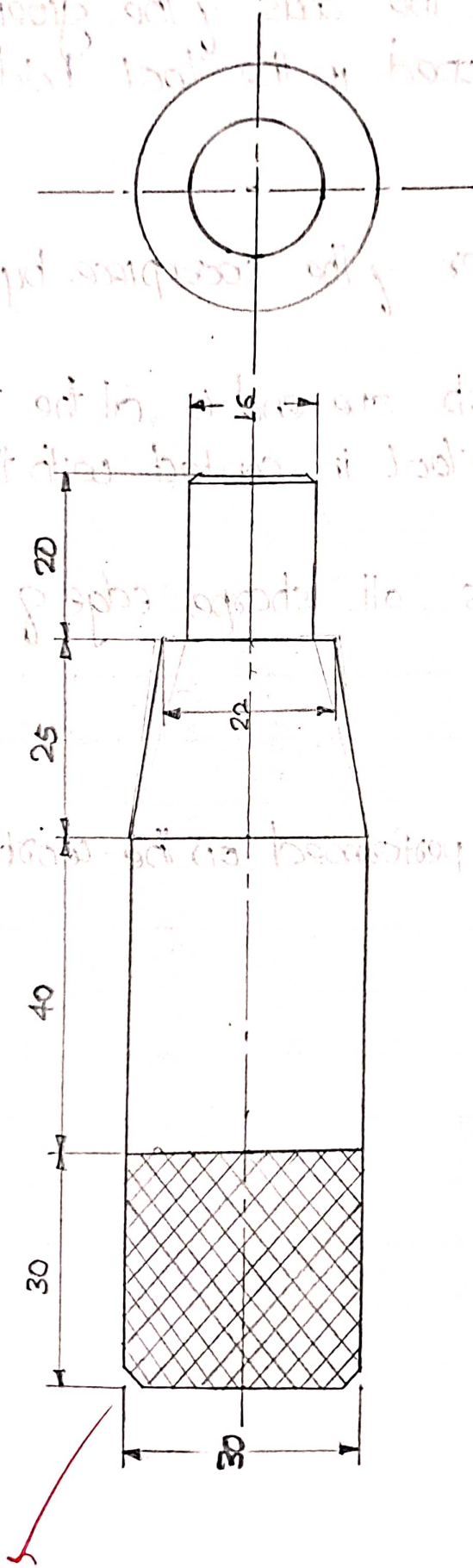
- (3) Reduce the diameter of the workpiece by turning process.
- (4) While doing the job one end is at the tool bit at 45° inclination and by bringing the tool in contact with the edge of the job.
- (5) This process resolves all sharp edge of the component.

RESULT ::

Step turning was performed on the workpiece.



TAPER TURNING OPERATION.



All dimensions are in "mm"

Scale 1:1

TAPER TURNING OPERATION.

When the diameter of a piece change uniformly from one end to the other, the piece is said to be tapered. Taper turning as a machining operation is the gradual reduction in diameter from one part of a cylindrical workpiece to another part. Taper can be either external or internal.

AIM:-

Test procedure to perform taper turning operation by compound rest swiveling method on the given cylindrical workpiece.

MATERIALS REQUIRED.

Lathe with standard accessories.

Workpiece: Mild steel round rod of diameter 30mm 30x115

TOOLS REQUIRED

Chuck key.

Cutting tool bit (H.S.S.).

Outside caliper.

Tool holder with key.

Spanner.

PRINCIPLE

Cutting taper on a lathe is common application. A number of methods are available for cutting taper on a lathe.

Large diameter - $D = 30 \text{ mm}$

Small diameter - $d = 22 \text{ mm}$

Layer length - $L = 25 \text{ mm}$

$$\tan \alpha = \frac{D-d}{2L}$$
$$= \frac{30-22}{2 \times 25}$$

$$\alpha = \tan^{-1} \frac{8}{50}$$

$$\alpha = 9^\circ$$

They are:

- ① Compound rest swelling method.
- ② Using Foam tool.
- ③ Tail stock offset method.
- ④ Paper attachment method.

These are the methods used for turning steps and short tapers. There is a circular base graduated in degree in which can be swiveled at any angle from the center of the lathe center. The amount of tapers in a workpiece is usually specified by the ratio of the difference in diameter of the taper to its length. This is termed as conicity and is designated by letter k .

$$\text{Conicity } k = (D-d)/L$$

Referring to the above figure BC drawn parallel to the axis and in the right triangle ABC.

PROCEDURE:

The workpiece is fixed in the tool post tightly and the center of headstock and tailstock is coincided with the center of headstock and tailstock. Facing and plane turning operation are performed to get the required diameter on the workpiece.

The compound rest is set on the required half taper angle and is locked by the cutting rod is adjusted to a fixed position for the best possible to the open hand.

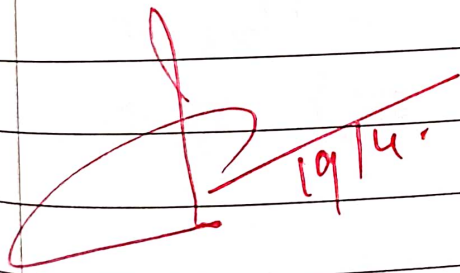
a fixed position. for the best possible to the open hand wheel and crossfeed. then the carriage is locked and first cut is made at the end of the cut, the tool is again cross feed. is given. for the next cut. cuts are repeated. piece is then removed. from the chuck and diameters. dimension. obtained. are noted.

PRECAUTIONS:

The workpiece should be fixed tight in the jaw. The power supply switched off before measuring diameters.

RESULT:

Power Turning was performed on the workpiece.

 19/4.